Adrià Mitjavila Ventura

Bioinformatician

Girona (Spain) amitjavilaventura.github.io amitjavilaventura

Bioinformatician with training in Molecular Genetics and Biotechnology and over 5 years of experience in omics data analysis, mainly transcriptomics and epigenomics.

Employment

Bioinformatician & **PhD candidate**, *Institut Josep Carreras* & *Institut Germans Trias i Pujol*, Badalona (Spain) *08/2021 - Present*

- Analysis and integration of small RNA-seq, RNA-seq, small RNA-seq data and 10x Xenium spatial transcriptomics data. Computational design of probes for 10x Xenium custom panel.
- Writing of scientific papers and reviews.
- Mentoring and outreach workshops adressed to students.
- Skills: bioinformatics and statistical analysis (R, Bash, Python), pipeline development (Snakemake), workflow and environment management (conda, docker, singularity), scientific writing, version control (git, GitHub), high performance computing (Slurm), outreach and mentoring.

Bioinformatician *Istituto Europeo di Oncologia*, Milan (Italy) *07/2019 - 08/2021*

- Analysis and integration of RNA-, ChIP- and ATAC-seq data in the context of colorectal cancer.
- Development of a Snakemake pipeline for ATAC-seq data processing and analysis.
- Skills: bioinformatics and statistical analysis (R, Bash, Python), pipeline development (Snakemake), environment management (conda, docker, singularity), scientific writing, version control (git, GitHub), high performance computing (PBS).

Master's thesis student & **Research assistant**, *Instituto de Biomedicina de Sevilla*, Seville (Spain) *09/2017 - 10/2018*

- Bioinformatic pre-processing and analysis of MNase-seq data.
- Designing of PCR primers, and performing and analysis of MNase-qPCR data.
- Skills: DNA extraction, MNase-qPCR, microbiological cultures, bioinformatics and statistical analysis (Bash, R, Past), scientific writing.

Undergraduate internship & **Bachelor's thesis student**, *Microbial*, Girona (Spain) *09/2016 - 07/2017*

- Analysis and processing of human intestinal biopsies, food and water samples for pathogen detection and food forensics.
- Statistical analysis of metagenomics data.
- Sample traceability and results documentation.
- Skills: DNA extraction, qPCR, molecular cloning, bacterial cultures, statistical analysis (SPSS), scientific writing.

Education

PhD in Bioinformatics, *Universitat Autònoma de Barcelona (UAB)*, Online *2022 - Present*

MSc. Bioinformatics and Biostatistics, *Universitat Oberta de Catalunya (UOC)*, Online *2018 - 2022*

MSc. Molecular Genetics and Biotechnology, *Universidad de Sevilla (US)*, Seville (Spain) *2017 - 2018*

BSc. Biotechnology, *Universitat de Girona (UdG)*, Girona (Spain) *2012 - 2017*

Skills

Languages: Catalan (*native*), Spanish (*bilingual*), English (*IELTS 7.5 - C1*), Italian (*basic*)

Bioinformatics: (small) RNA-/ChIP-/ATAC-/MNase-seq, single-cell RNA-seq, 10x Xenium

Informatics & Statistics: R/Bioconductor, R Markdown, Bash, AWK, Python, Snakemake, Conda, Docker, Singularity, HPC (PBS & Slurm), SPSS, git, GitHub.

Publications

Mitjavila-Ventura & Vavouri. **Mechanisms seeding the emergence of new piRNAs**. *Trends in Genetics*. 2026. doi: 10.1016/j.tig.2026.04.004

Casas, Mitjavila-Ventura *et al*. **Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse**. *EMBO Journal*. 2025. DOI: 10.1038/s44318-025-00475-4.

Begley *et al*. **Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration**. *RNA Biology*. 2021. DOI: 10.1080/15476286.2020.1845504.